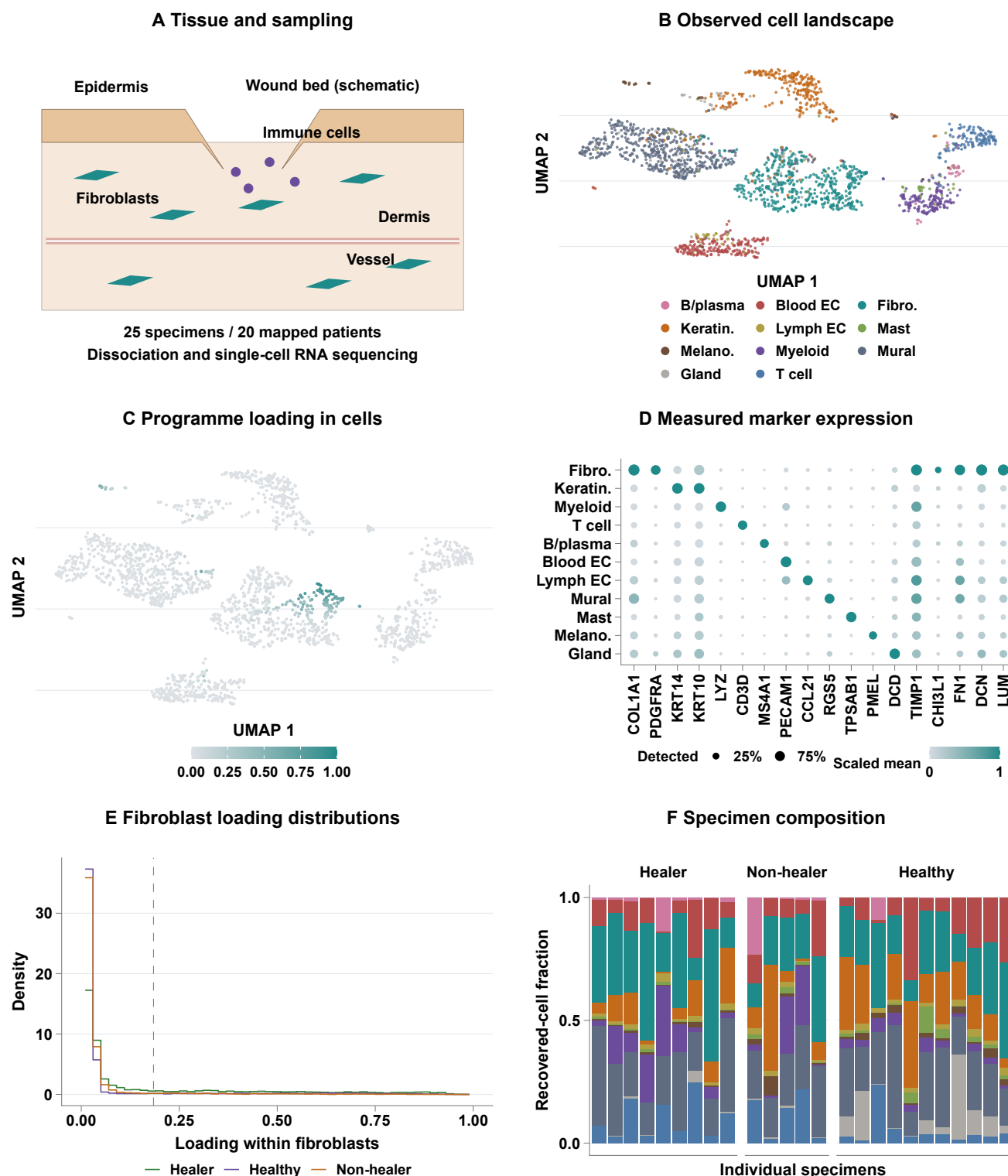
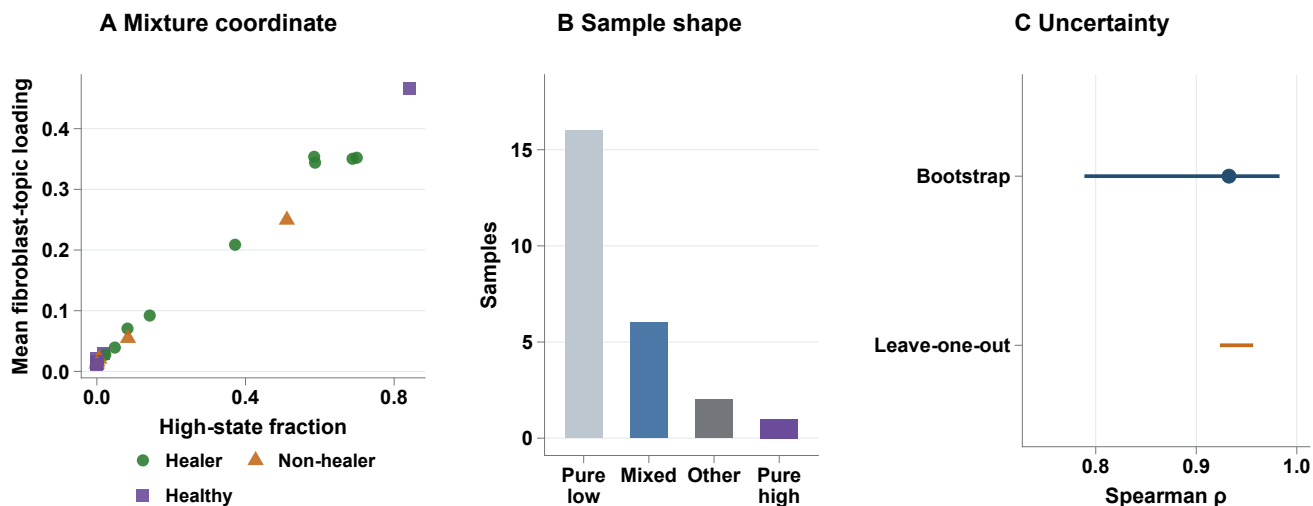


Figure 1

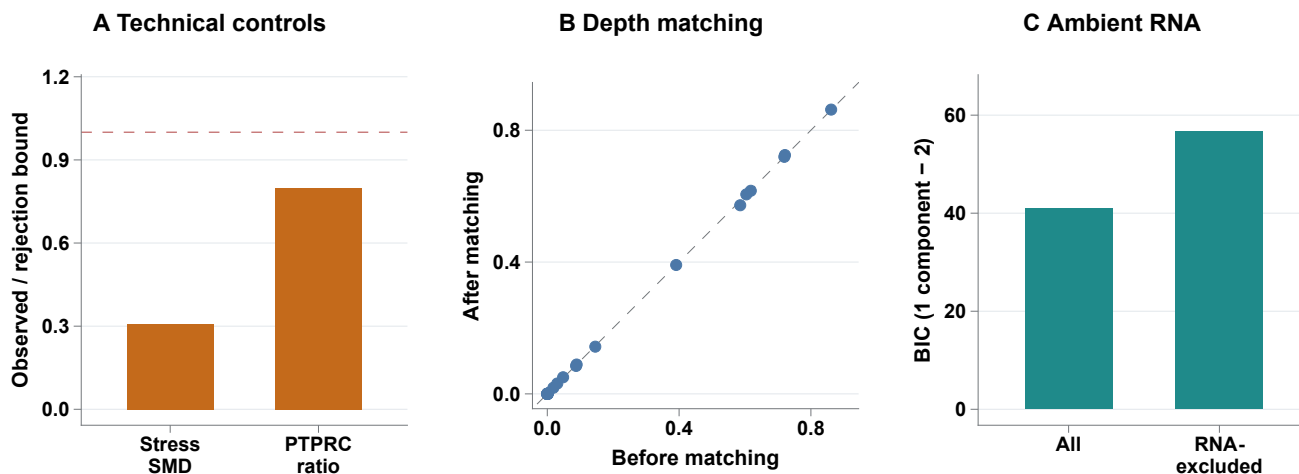


Observed study material and expression context. (A) Conceptual skin anatomy and sampling, not a tissue image or inferred spatial map. (B) A uniform 2,000-cell subset of the 76,461-cell UMAP, colored by marker-derived lineage. Fibro., fibroblast; EC, endothelial cell; mural, pericyte/smooth muscle. (C) The identical cells colored by frozen programme loading; the map and colors do not define new cell types. (D) Raw expression across all retained cells: dot size is the detection fraction and color the gene-wise scaled mean log-normalized expression. Annotation markers provide descriptive, not independent, support for labels. (E) All 19,410 fibroblast loadings in fixed-width histograms, with the existing threshold marked; healthy participants have no healing outcome. (F) Recovered-cell composition in each of 25 specimens, grouped by source outcome or non-diabetic status. Colors match B; columns are specimens, including repeated patients. Inference uses the relevant full data and patient units, not the display subsample.

Figure 2

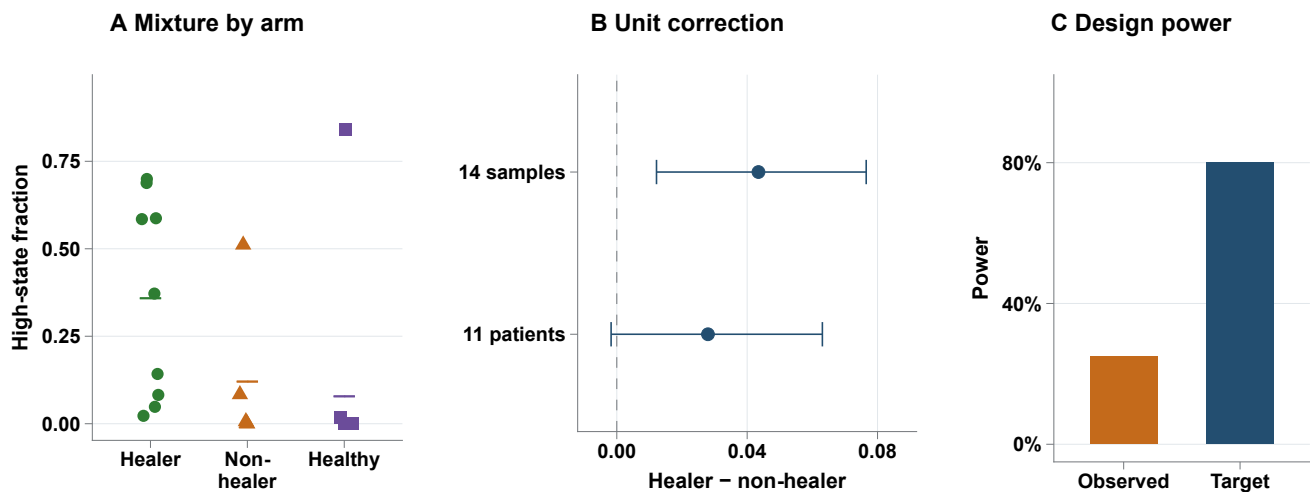


Association between mean topic loading within fibroblasts and the high-state fraction. (A) Each point is one of 25 GSE165816 specimens: 9 healing (green circles), 5 non-healing (orange triangles) and 11 healthy non-diabetic (purple squares). The high-state fraction uses the pooled cut 0.1842958; Spearman $\rho=0.933$. (B) Sample-shape counts use within-sample loading quantiles: pure low requires the 90th percentile at most 0.15, pure high the 10th percentile at least 0.10, and mixed the 10th percentile at most 0.05 and 90th percentile at least 0.40; other samples form the remaining category. (C) The dot is the observed correlation, the blue segment is the 10,000-draw sample-bootstrap 95% interval (0.789–0.983), and the orange segment is the leave-one-sample-out range (0.924–0.957). Both summaries use the same cells; the association does not by itself prove stable within-state intensity or a healing effect.

Figure 3

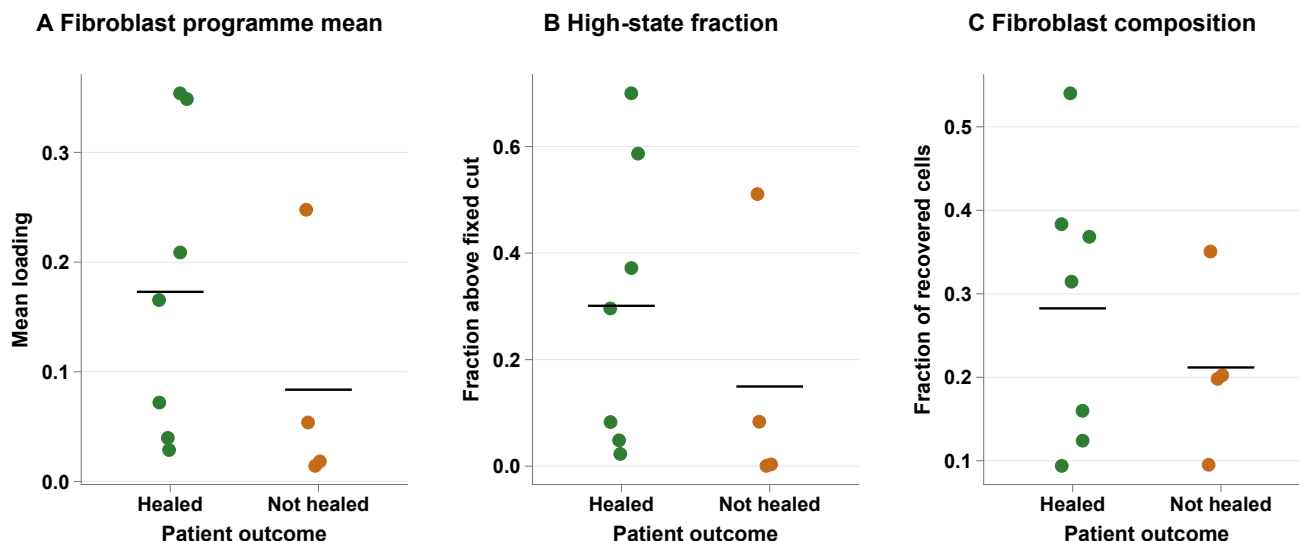
Technical and immune-transcript exclusion controls in the discovery cohort. (A) Dissociation standardized mean difference (SMD, 0.246) and the high-state/low-state PTPRC-positive rate ratio (1.60) are divided by rejection bounds 0.8 and 2.0; the dashed line is one. The rounded state cut is 0.184. (B) Each point is one of 25 samples after restricting to 14,561 fibroblasts eligible for panel-count matching to 986 counts per cell. Reprojection preserves sample fractions ($\rho=0.984$) and 99.5% of cell labels; the dashed line is identity. (C) Bars are BIC for one Gaussian minus BIC for two Gaussians fitted to sample high-state fractions, using 24 matched samples before and after exclusion. Positive values favour two components. Exclusion retains 11,262 of 19,410 fibroblasts overall, and matched sample fractions correlate at $\rho=0.989$. This control measures invariance to the exclusion rule; it does not identify every source of ambient RNA or tissue-composition dependence.

Figure 4

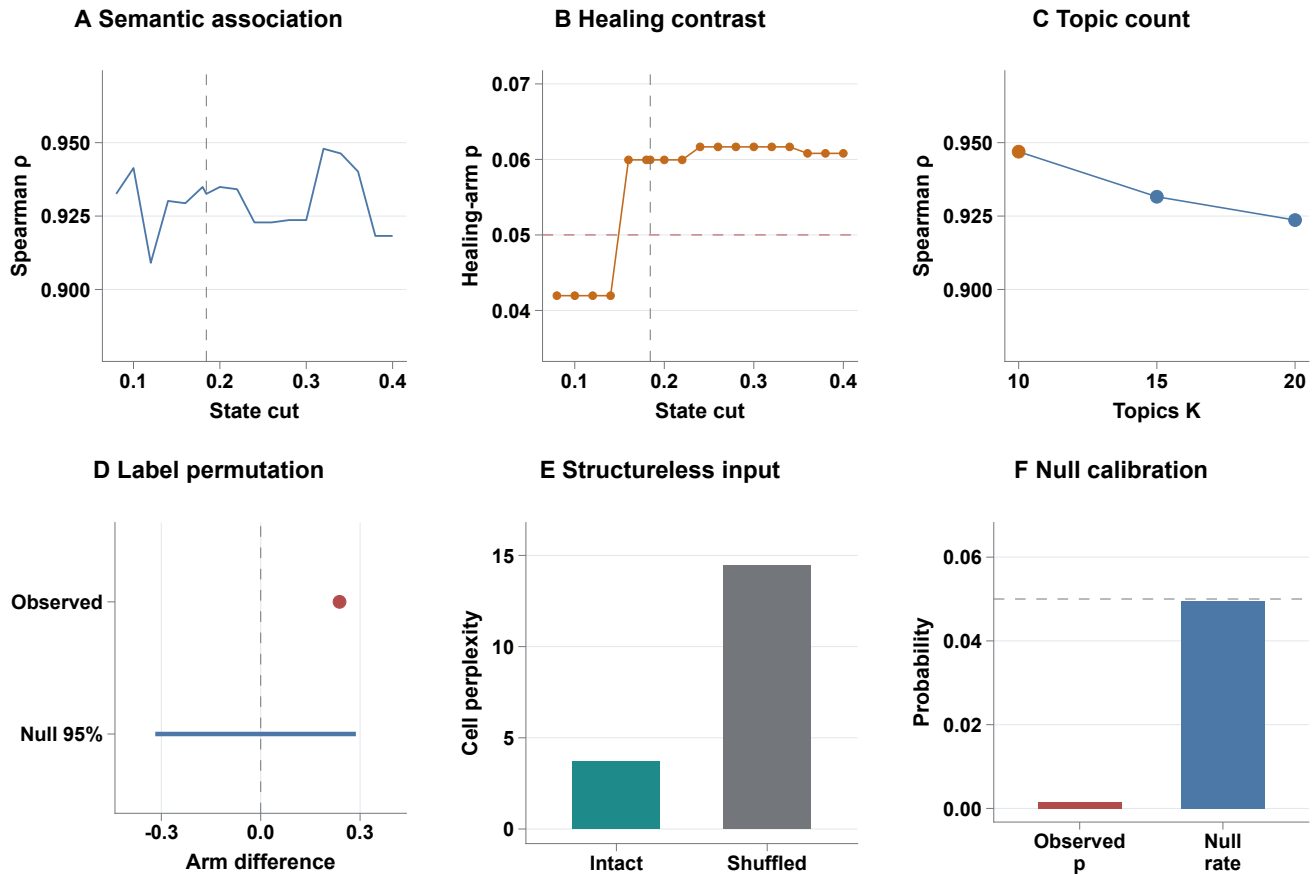


State abundance, analysis unit and design power. (A) Symbols show high-state fractions for 9 healing, 5 non-healing and 11 healthy specimens; horizontal segments mark arm means. Green circles, orange triangles and purple squares identify these groups. The largest fraction is 0.841 in healthy non-diabetic foot skin, showing that the state is not exclusive to healing ulcers. (B) Dots and within-arm bootstrap 95% intervals show the healing-minus-non-healing difference in all-cell mean fibroblast-associated topic loading for 14 specimens and 11 mapped patients. The patient analysis resamples 7 healing and 4 non-healing patients in 30,000 draws. Its readout differs from the fibroblast high-state fraction in A; the vertical dashed line is zero. (C) Gaussian location-shift simulation with a two-sided Mann–Whitney test estimates 24.9% power for the separate 9-versus-5 high-state-fraction design, against an 80% target. Its minimum detectable difference is 0.519. These are design sensitivities, not effect-exclusion bounds or clinical minimum important differences.

Figure 5

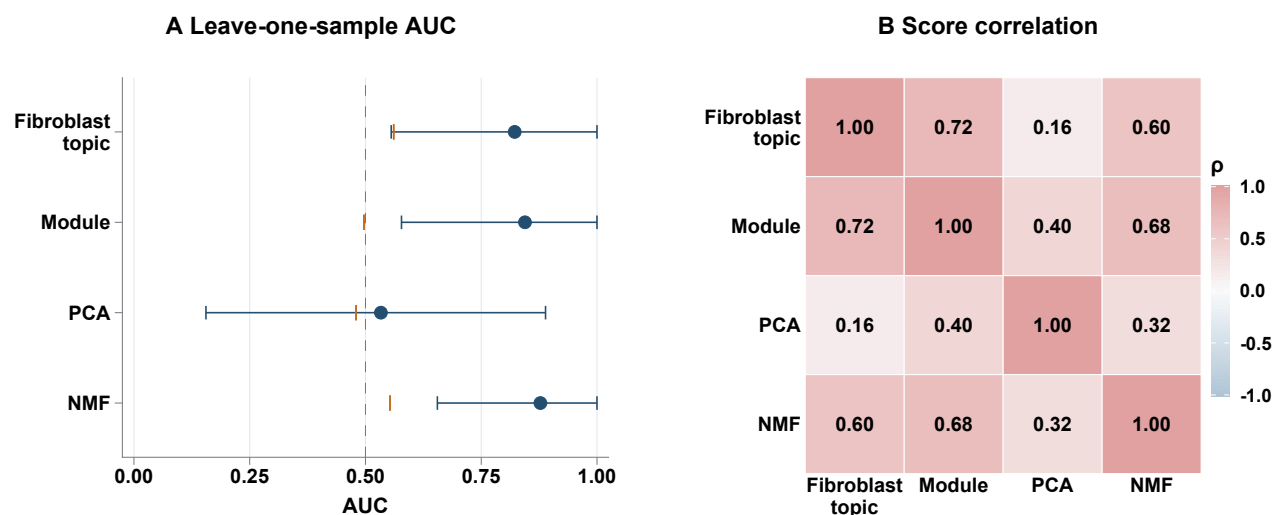


Patient-unit evaluation of three distinct tissue readouts. Each point represents one of 7 healed or 4 non-healed DFU patients; horizontal bars show group means. Repeated specimens are pooled within patient using the relevant cell counts, with no outcome-based change to the model or threshold. (A) Mean loading among fibroblasts: healed-minus-non-healed difference 0.0892, exact permutation $p=0.3121$. (B) Fraction of fibroblasts above the fixed threshold: difference 0.1514, $p=0.3697$. (C) Fibroblast fraction among all recovered cells: difference 0.0708, $p=0.4697$. All 330 patient-label allocations enter each two-sided test without add-one correction; BH adjustment over the three tests gives $q=0.4697$ for each. Patient-bootstrap intervals and weighting sensitivities are reported separately. Points are independent patient summaries for the outcome comparison, not cell-level replicates.

Figure 6

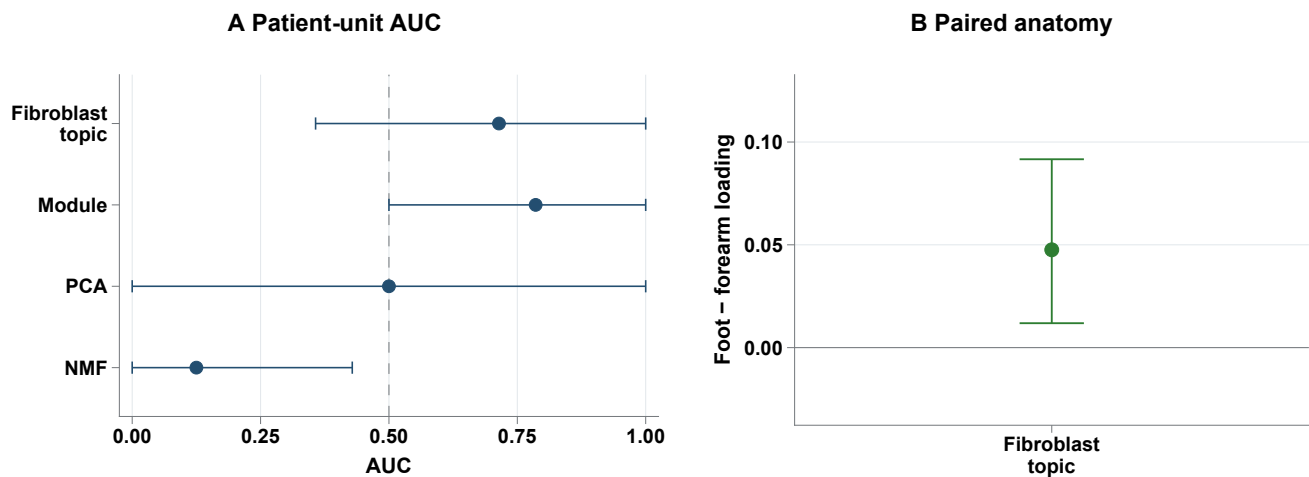
Threshold, capacity and negative-control analyses. (A) Each point gives the Spearman correlation between mean fibroblast loading and high-state fraction across 25 discovery specimens at one of 18 cuts from 0.08 to 0.40. The vertical dashed line marks the pooled fitted cut, 0.1842958. (B) Each point gives the two-sided Mann–Whitney healing-arm p value for 9 healing versus 5 non-healing specimens at the same cuts. The vertical line marks the fitted cut and the horizontal line marks $p=0.05$; 4 of 18 values fall below that level. (C) Correlations for $K=10$, 15 and 20 after tracking the TIMP1-carrying topic; orange marks $K=10$, where the highest fraction is no longer in healthy skin. (D) The observed high-state-fraction arm difference is compared with the central 95% of 20,000 label permutations ($p=0.1358$). (E) Gene-column shuffling of 6,000 cells raises mean cell perplexity from 3.68 to 14.43 out of 15, violating the concentration criterion. (F) The anatomical sample-level control compares 33 foot and 11 forearm specimens: observed $p=0.0015$, with 5.0% of permuted comparisons below 0.05. This calibration is distinct from the paired 10-patient anatomy analysis and does not establish pipeline-wide error control.

Figure 7



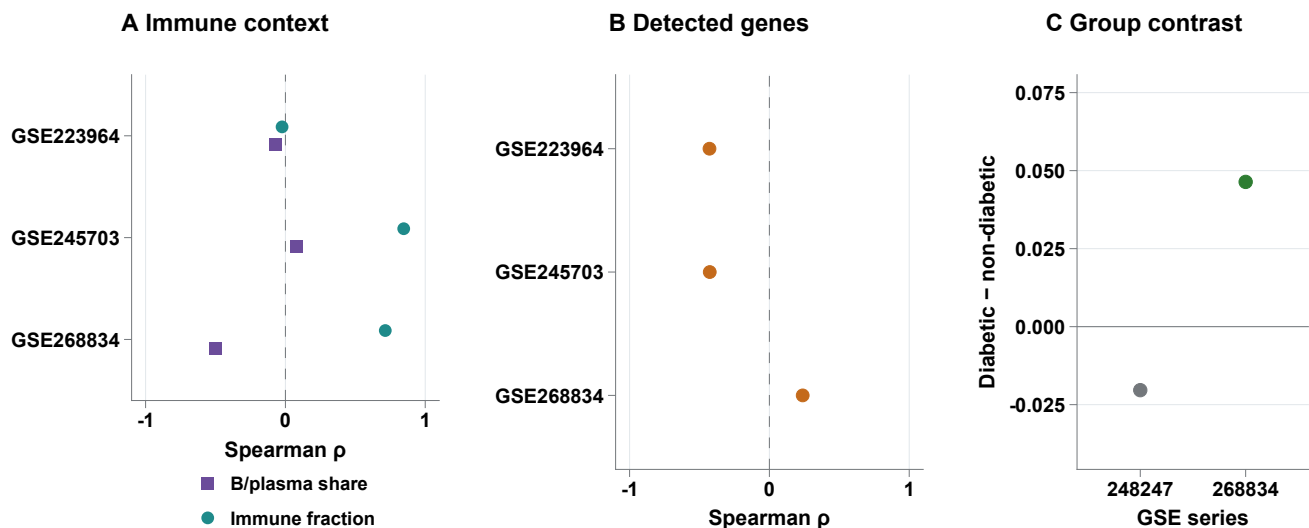
Sample-unit representation comparisons in 14 DFU specimens (9 healing, 5 non-healing). (A) Points are leave-one-sample-out areas under the ROC curve (AUCs); horizontal bars are percentile 95% intervals from 4,000 arm-stratified bootstrap draws of fixed out-of-fold scores, with common resampling indices across methods. Orange ticks mark the permutation-null means after rerunning outcome-dependent fitting and orientation. The dashed line at AUC=0.5 is a visual reference, not the fitted permutation null. Fibroblast topic denotes the frozen all-cell mean, Module the specified fibro-inflammatory score, PCA principal component analysis, and NMF nonnegative matrix factorization. No paired AUC-difference interval excludes zero. (B) Cells contain Spearman correlations of the 14 sample-level readouts; the colour scale and printed values encode the same statistic. These exploratory comparisons do not establish a clinical method ranking.

Figure 8



Patient-unit discrimination and paired anatomical sensitivity. (A) Points are leave-one-patient-out AUCs for 7 healing and 4 non-healing patients after cell-count-weighted collapse of repeated specimens. Bars are 95% intervals from 4,000 arm-stratified patient bootstrap draws of fixed out-of-fold scores; the dashed line marks AUC=0.5. Fibroblast topic, Module, PCA and NMF denote the frozen topic mean, fibro-inflammatory module, principal component and non-negative matrix factorization readouts. (B) The point is the mean all-cell fibroblast-associated topic loading difference, foot minus forearm, across 10 paired patients; the bar is a 4,000-draw paired-patient bootstrap 95% interval and the horizontal line is zero. The exhaustive sign-flip test with add-one correction gives $p=0.0107$ and Benjamini–Hochberg $q=0.0402$ across 15 topics. Both panels use the discovery source and are not independent clinical validation.

Figure 9



Frozen external projections under mutually exclusive lineage gates. Correlations use 8 GSE223964, 12 GSE245703 and 8 GSE268834 specimens. (A) Each point is a cohort-level Spearman correlation between the sample mean fibroblast-associated topic loading within fibroblasts and either immune fraction among all QC cells (teal circles) or B/plasma share within immune cells (purple squares). (B) Orange points correlate the same topic mean with median detected genes per cell. The vertical dashed lines in A and B mark zero. (C) Points show diabetic-minus-non-diabetic differences in the sample fibroblast mean: +0.0464 in GSE268834 (descriptive permutation $p=0.2812$) and -0.0204 in GSE248247 ($p=1.0$). GSE248247 contains one specimen per group and 358 QC cells, so its correlations are not estimated. No panel displays confidence intervals. These cohorts lack qualifying patient-level healing endpoints and are not pooled for prognosis inference.